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Software Requirements Specification

for

HealthyCooking System

SRS
Version 1.0

26th of October 2025

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1. Introduction

The section will address the topics related to the SRS document for the first version of the HealthyCooking System project. It lays out everything in detail, including purpose, docuument conventions, intended audience, and procduct scope.

1.1 Purpose

The main goal of this document is to state, describe, and give all the requirements of HealthyCooking software system. The proposed system is described in detail in terms of its functionality, non-functionality, external interface, and more in this SRS document.

This document mainly concentrates on specifying and systemizing the requirements of a health-oriented software system which can assist the users in managing their daily nutrition and healthy meal planning. The most essential aim of the project is to create a healthy and disease-fighting diet in a balanced way, as advised by the WHO [2].

The SRS seeks to bring all project stakeholders to a common understanding of the intended behavior and use of the system. We do this phase to minimize ambiguity. Also, it helps establish the basis of the subsequent design, and testing phases.

Thus, this is a formal agreement between a supervisor, stakeholders and the development team of HealthyCooking project requirements. So, they are all clearly defined and understood by all.

1.2 Document Conventions

This document follows the IEEE standards for Software Requirement Specification documents to maintain consistency, clarity, and completeness throughout all sections. The requirements are written in a simple and clear format without using any special highlighting. Each requirement is given its own priority level separately.

1.3 Intended Audience and Reading Suggestions

The document is organized into sections that describe the overall system, specific requirements, and supporting information. Readers are encouraged to begin with the overview sections to understand the project scope, and then proceed to the detailed requirements according to their role and interest.

The development team refers to this SRS as the main reference during all development activities to ensure that all functional and non-functional requirements are implemented.

This document contains the intention of the system, project deliverables, and project scope which need to be known by the stakeholders and evaluators before authorizing or evaluating the system.

The Project Supervisor is responsible for checking the SRS to ensure it is up to standard as a document related to the academic and professional world, as well as commenting and estimating the Project Work and Guide for practical evaluation.

1.4 Product Scope

This section defines the general scope of the HealthyCooking Software SRS document. It describes the key requirements that define the system's core services and identifies the main constraints that may affect performance, usability, and data accuracy.

HealthyCooking will help effectively plan, find, and manage healthy meals. The system will offer multiple intelligent features that help users explore recipes, track their nutrition, and get customized meal recommendations [3]. But the system will not provide for diagnostics, recommend specific diets for health conditions, or substitute for dietary or clinical advice.

When data is handled securely and user preference is accurately analyzed, it will ensure privacy, reliability and trust. The document includes all the functional and non-functional requirements of the system. It also works as a systematic reference that limits the project and ensures that everyone involved knows the purpose and scope clearly.

The project aims to promote healthy eating behavior and make nutritional management easier and more pleasurable for end-users. The system aims to enhance the well-being of customers by presenting them with accurate suggestions, personalized dietary plans, and an authentic database of healthy recipes. This fulfils the primary criteria set for the project and guarantees effectiveness, dependability and user satisfaction.

Role	Functionalities Provided
Admin	Allows the admin to: • Oversee and maintain the nutritional and recipe database. • Examine and approve user-submitted recipes. • Monitor system performance and ensure data accuracy. • Manage technical issues and user feedback while maintaining system security. • Verify content quality and conformity with health regulations.
User	Allows the user to: • Create and manage a personal profile with dietary preferences and health goals. • Search, filter, and discover healthy meals by ingredients, meal type, or calorie count. • Create customized weekly meal plans and receive helpful recommendations. • Track daily nutritional intake and monitor progress over time • Save favorite recipes and share comments or reviews with the community.

Table 1: System roles

1.5 References (IEEE Style)

This section lists the references used in preparing the HealthyCooking Software Requirements Specification (SRS). These references include nutritional standards, health guidelines, and reliable online sources that support the system's features and functionality.

[1] OpenAI, ChatGPT, https://chat.openai.com.

[2] World Health Organization (WHO), "Healthy diet," [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/healthy-diet. [Accessed: Oct. 25, 2025].

[3] U.S. Department of Agriculture (USDA), "MyPlate Guidelines," Center for Nutrition Policy and Promotion, [Online]. Available: https://www.myplate.gov/. [Accessed: Oct. 25, 2025].

2. Overall Description

This section provides an overview of the HealthyCooking application. It outlines the background of the system, the constraints that may affect its operation, and its interactions with external components from the product perspective. The product functions subsection highlights the key features available to both administrators and end users of HealthyCooking. The system's main characteristics that support healthy eating practices include a recipe and nutrition tracker, along with an intelligent meal planner.

2.1 Product Perspective

HealthyCooking is a standalone software product enabling users to receive personalized nutrition information and effectively plan healthy meals.

The system connects to a recipe database and uses smart algorithms to create personalized recommendations based on user preferences and dietary objectives. Healthy eating advice may come from outside sources, such as the recommendations of reputable health organizations.

HealthyCooking focuses on simplicity and support in healthy eating through features to track nutrition, plan meals and explore recipes. This was designed to be scalable and flexible enough to enable future enhancements with additional dietary modules and sophisticated analytics.

The overall view of the system and its connection with end-users and external elements are represented in the following diagram, this diagram illustrates the relationship between the HealthyCooking system, users, admin, and the recipe database as external entities.

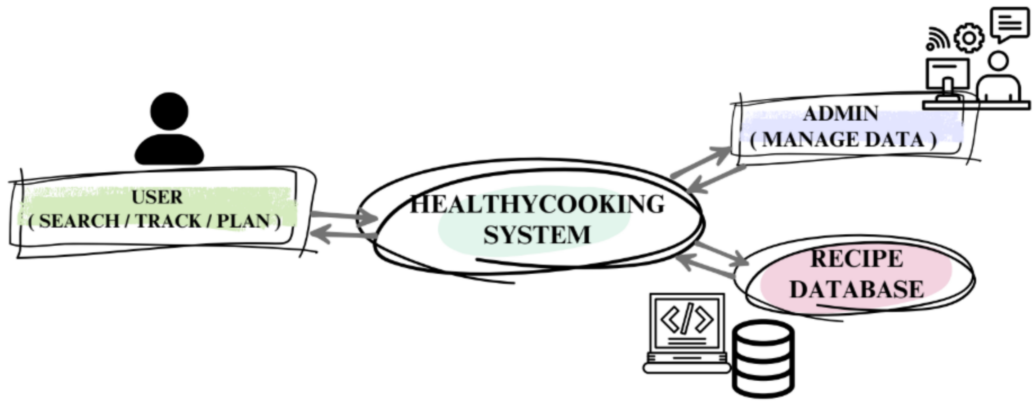

Figure 1: HealthyCooking system context diagram

2.2 Product Functions

HealthyCooking System uses intelligent recommendations to assist users find, prepare, and track meals in order to encourage healthier eating habits. Each of the key modules that make up the basic system functions servers to a particular set of user requirements.

1. Recipe Discovery
   - Allows you to filter or search through hundreds of recipes according to flavor profiles or variety of dietary restrictions or prep time, types of cuisine and more filters could be added in the future.
   - Lets you filter your search based on exactly what you're craving and actually need form a meal. For example: high protein, low-carb, gluten free, under 30 mins and other options. In addition, filters can be used to refine the choices of meals (e.g. high protein, low carb, gluten-free, under 30 mins, etc.) and multiple filters can be applied simultaneously (e.g. high protein + low calorie meal, etc.).
2. Personalized Recommendations
   - Because all meal recommendation are based off your preferences, dietary goals and/or restrictions the meal planners utilizes AI and machine learning algorithms to align for you more specified recommendations.
   - Plus improve recommendations based on user content in the food portal with focus, thoughts, and actions to like and save recipes.
3. Smart Ingredient Analyzer
   - Allows users to discover meal options that they could make based on what they already have and to enter their available ingredients.

- If certain ingredients are missing, suggestions of recommended foods can be made.

4. Weekly Full Meal Planner
   - Creates every week a balanced meal plans that takes user goals, dietary restrictions, and caloric limits into consideration.
   - Provides the user with full control and flexibility to modify any said plans.
5. Integration with Nutritional Databases
   - The system is going to be integrated with a public databases for nutritional information (e.g., USDA FoodData Central) to calculate and show the calorie and macronutrient profiles of each meal.
   - Live updates occur when meals are modified, and nutritional information may automatically be computed.
6. Recipe and Community Sharing
   - Allow individuals a place to exchange personalized recipes and experiences.
   - Provides a commenting and rating/feedback to encourage community involvement.
7. User Profile Management
   - Provides users the ability to set and maintain profiles that contain dietary objectives/allergies and personal preference information.
   - Provides data privacy and secure processing of data by encrypting user information stored.
8. Analytics Dashboard
   - Provides visual reports and graphs of nutrient distribution, daily, and weekly calories of information, and a summary of user progress over time.Helps customers in keeping track of their nutritional goals.
9. Reminders and Notifications
   - Notifies users automatically when meals, groceries, or daily progress reports are due.
   - Keeps users consistent and motivated..

2.3 User Classes and Characteristics

The HealthCooking system caters to four unique user types: regular users, supermarket staff users, nutrition professional users, and system integrations users. Each type is intended for different purposes, has different technical knowledge and skills, and each has different access or privileges on the system. The majority of the work will be done by the regular user, who will register for the system, sign in, manage their own profile, view the recipes, create a meal plan, review nutrition information, and make decisions about their recommendations. The administrator user type will have the typical responsibilities of administrative tasks, such as managing user database, validating recipes, answering to users, and general system updates and maintenance and support.They possess a reasonable amount of technical expertise. Upcoming updates might include in app Nutrition Specialists, qualified professionals, that are able to confirm nutrition information, provide verified health content, and deliver expert recommendations. In addition, the most skilled group, system developers, will handle development, testing, and deployment to ensure scalability and reliability.

2.4 Operating Environment

HealthyCooking application operate on iOS and Android software and hardware environment. It needs an internet connection to connect with the server and database to store the recipes and user preferences. The application also depends on temporary local storage to keep some data available when the device is offline.

2.5 Design and Implementation Constraints

The overall architecture and development of the system will be influenced by several constraints. These include both organizational and product-related factors that define how the HealthyCooking application is developed and maintained. In the following table, we discuss the constraints in detail.

Constraints	Description
Regulatory Policies	The application must comply with general data privacy and security practices required for mobile applications. It should protect user accounts, stored preferences, and recipe data.
Hardware Limitations	The system must perform efficiently on typical smartphones and tablets without relying on high-end hardware or large storage capacity.
DataBase Constraint	A real-time database will be used for storing and retrieving user and recipe data to ensure synchronization, consistency, and smooth interaction across all modules.
Security Constraints	Appropriate authentication, encryption, and access control mechanisms must be implemented to protect user data and ensure system reliability against unauthorized access or misuse.
Budget Constraint	Development should remain within the available resources to ensure cost-effectiveness and efficient use of time and effort.

Table 2: Design and implementation constraints

2.6 User Documentation

This system will be accompanied by concise user documentation to guide users in installing, configuring, and using the HealthyCooking application, the following documentation will be provided:

PDF User Manual: to describes basic features such as recipe search, meal planning, and nutritional tracking, with screenshots of the application interfaces.

Quick Start Guide: one-page summary of the main actions (search - plan - track).

In-App Help Tooltips: short contextual hints displayed beside icons and buttons.

FAQ Section: clarifies common issues, such as managing allergies, editing preferences, or exporting weekly plans.

All user documentation will be written in English and Arabic, and stored in the project repository as part of the prototype deliverables.

2.7 Assumptions and Dependencies

We focused on creating our Healthy Cooking app with many ideas, including the fact that internet access is necessary for the user as it's required to browse and search the app, receive AI suggestions, organize, edit, and save meal plans and meals It's also important that the device has sufficient storage space to run the system, receive the latest updates, and ensure smooth and efficient use. We guarantee that our app's recipe information is accurate, reliable, and trustworthy. We assume users accurately enter their personal data, such as any food allergies, or add recipes to their favorites so our program can suggest suitable options The system depends on different external services. The application requires a cloud database to store recipes and user information. It also depends on AI services to generate smart suggestions. In addition, the app needs access to Google Play Store and Apple App Store for installation and updates. If any of these services stop working, some features of the application may be affected.

3. External Interface Requirements

3.1 User Interfaces

This section outlines the logical characteristics and optimization aspects of the interface between our HealthyCooking application and its users. Our system will be designed with a simple, clear, and user-friendly interfaces that is also quick to define and responsive. This will help users easily plan and manage their healthy meals. Our interfaces prioritize quick access, speed, and ease of use because we care that all users of the application receive interaction regardless of their skill level or experience. This includes interaction with AI which is very helpful as it provides users with suggestions and recipes using specific ingredients or other elements and displays a wide variety of recipes.

Login Interface:

This interface gives the program's users access to their accounts in complete security. The system requires our program to verify the accuracy of the entered data, as well as to clarify and approve it before granting access to the saved data, such as healthy meal planning, preferences, etc.

Input Field	Description	Type	Requirement
Email	Registered user email	Text	Required – must be unique
Password	User password	Encrypted Text	Required – minimum 8 characters including letters, numbers, and symbols

Table 3: Login interface components

for the table above, if the data entered by the user is incorrect, an error message will be given and displayed for either the email or the password: "The email or password is incorrect here." It will also give the option "I forgot my password" in case the user forgets and to reset the data.

Forgot Password Interface:

The program allows users to reset their passwords through a single method: verifying their email address. If the email address is correct, a confirmation message will appear: "Password successfully renewed and updated."

Input Field	Data Type	description
Email	Text	Used to identify account
New Password	Encrypted Text	Creates a new password

Table 4: Forget passwords interface components

Registration Interface:

The program interface allows a new user who intends to register to create an account and save specific data such as dietary preferences, excluded ingredients, etc.

Field	Data Type
Full Name	Text
Email	Text
Password	Encrypted Text

Table 5: Registration interface components

After confirming and accepting the entered registration, the system displays the message "Account created successfully." If the email address already exists, it will display "This email address is already registered".

Home Interface:

The program gives the user the main screen with quick access to recipes, filters, approved recommendation and mor, Here, our AI engine automatically suggests recipes in our program based on the user's previous searches in the program, the type of cuisine, and their dietary goals.

Component	Function
Search box	The search box allows searching by ingredient, recipe name, or cuisine type.
Filter	Our program filters recipes by cuisine type, specific dish type, and preparation time in minutes or hours.
Recipe	Displays recipe image, title, and calorie info
Navigation Bar	Provides links to Home, Planner, Favorites, and Profile

Table 6: Home interface components

This interface serves as the main basis for viewing and discovering recipes with quality and efficiency

Profile and Settings Interface:

The program allows users to refresh and update their preferences, personal information, and dietary requirements The interface confirms saved updates with a "Profile updated successfully" message.

Items	Function	Format	Type
Name	Displays or edits user full name	Required	Text
Excluded Ingredients	Allow users exclude unwanted ingredients	Required	Text
Save Settings	Confirms and saves updates	Optional	Button

Table 7: Profile and settings interface components

Recipe Details Interface:

It provides complete information about our recipes, including the steps and ingredients needed for the recipe and the nutritional value of the meal. It also allows the user to add them to their favorites or to their recipe program.

Component	Function	Type
Recipe name	It provides the name of the recipe	Text
Ingredients	This Lists all required ingredients	Text
Steps	Give preparation instructions	Text
Nutrition Info	Shows calories and nutrients	Number/text
Save Button	Adds to favorites	Button
Add to Planner	Schedules meal for a chosen day	Button

Table 8: Recipe details interface components

If the recipe is already in the program in the user's favorites, it will give him a message: "The recipe has already been saved." The AI system analyzes the recipe specified by the user and provides or suggests alternative ingredients or gives the user similar recipes based on the user's preferences and the items available here.

Meal Planner Interface:

It allows users to plan meals throughout the week and track their food intake.

Feature	Function	Type
Calendar View	Provides days of the week for meal scheduling	Text
Recipe List	Adds recipes to each meal	Text
Total Calories	Calculates daily nutritional	Data
Meal Type Selector	Selects (Breakfast, Lunch, Dinner)	Text
Add Meal	Add new recipe to planner	Button
Remove meal	Remove a selected recipe	Button

Table 9: Meal planner interface components

Users of the program can modify or delete selected or planned meals Our system automatically calculates the total calories immediately after any change.

Favorites Interface:

Here Display recipes saved by the user for quick access to recipes.

Item	Function
Remove Button	Deletes recipe from favorites
Open Button	Opens recipe details page
Recipe List	Displays all saved recipes we have

Table 10: Favorites interface components

Our design for the Healthy Cooking application interfaces focuses on ease of use in all aspects, including performance, simplicity, clarity, and accessibility. Text should be clear and simple; fonts ensure easy readability for everyone. The application provides clear, user-friendly, and comprehensive alerts that explain incorrect actions, and errors, enabling users to understand, correct, and modify them. It also allows users to complete important tasks in just few clicks. It is compatible with all devices, whether phones, tablets, or personal computers, across all versions, allowing the user to access them.

3.2 Hardware Interfaces

HealthyCooking application is used on smartphones and does not need any specialized hardware. It is an app targeting Android and iOS devices. The touchscreen on the phone allows users to use the app to browse the recipe searching, nutrition calculations, and AI suggestions.

The device must be large enough in terms of storage of the app and some data saved in order to perform well. There is also need of a stable internet connection (Wi-Fi or 4G or 5G) to view recipes and AI features. The application does not require any devices or equipment, and this is why it is simple to utilize it by any person.

Data format / protocol	Description	Interface Name
SQL / JSON API	Internal database storing recipe metadata, ingredients, and nutritional information.	Recipe Database Module
HTTPS – REST / JSON	External public API (e.g., USDA FoodData Central) used to fetch nutrient data per ingredient.	Nutrition API
HTTPS – OAuth 2.0 / JWT	Manages user login and session tokens.	Authentication Service
Local Log File → CSV	Collects anonymized user activity for performance and usability metrics.	Analytics Module
Firebase Cloud Messaging	Sends reminders about planned meals or grocery alerts.	Notification Service

Table 11: Software interface components

3.3 Software Interfaces

The HealthyCooking application interacts with a cloud database for storing user profiles, recipes, and meal plans. It utilizes AI service APIs to produce meal recommendations and depends on the device's operating system (Android / iOS) for notification and storage access. Data is securely transmitted between the app and the server via HTTPS, guaranteeing accuracy and confidentiality in every interaction

The HealthyCooking system logical software interfaces include the following components shown in the table below:

3.4 Communications Interfaces

The HealthyCooking application requires a reliable and secure connection between the user interface and the server. All data transmissions, including user authentication, recipe sharing, and nutrition updates are encrypted and transferred via the HTTPS protocol to ensure data security and privacy. Communication strictly follows secure web standards to prevent unauthorized access or data breaches. In case of temporary network interruptions, the system includes an automatic reconnection mechanism to maintain stable operation and data synchronization.

4. System Features

This section presents the major functional requirements for the HealtyCooking application by use case figures.

4.1 System Feature 1

Sign-up

4.1.1 Description and Priority

Sign-up allows users to create an account in the HealthyCooking system to be able to access all the features in it. The following figure illustrates the use-case of this feature. Priority of this feature is high. Benefit: 9 - Penalty: 7 - Cost: 5- Risk: 3.

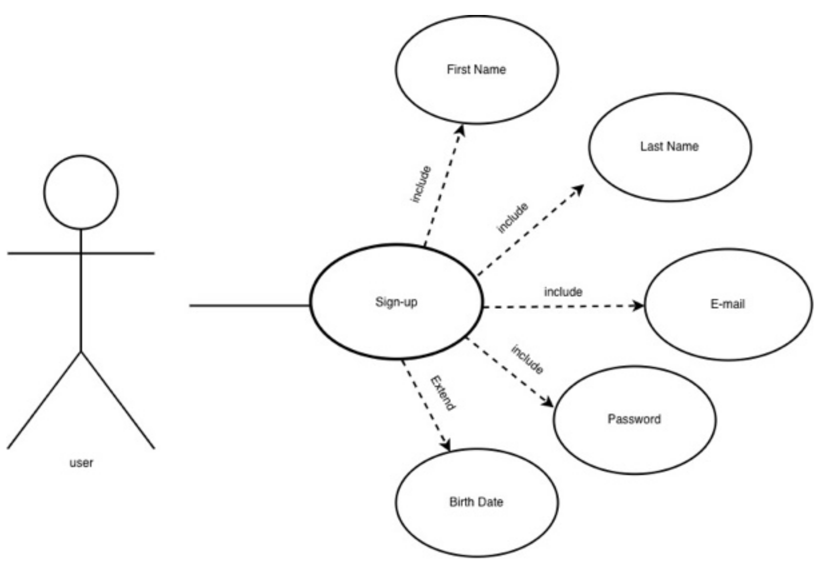

Figure 2: use case of the sign-up feature

4.1.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Sign-up</th></tr>
<tr><td>Goal</td><td>Enable user to register in the system.</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>Database</td></tr>
<tr><td>Pre-condition</td><td>User downloads the application</td></tr>
<tr><td>Post-condition</td><td>User register in system successfully.</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to sign-up</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “sign-up” button.</td></tr>
<tr><td>2</td><td>System displays first name, last name, E-mail, password, and birth date filling interface.</td></tr>
<tr><td>3</td><td>User enters first name, last name, E-mail, password, and birth date.</td></tr>
<tr><td>4</td><td>System checks the validity of the E-mail</td></tr>
<tr><td>5</td><td>System saves user inputs in the database and shows a message of successful registration.</td></tr>
<tr><td>6</td><td>System directs user to the home interface.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>4.1</td><td>The E-mail is not valid, system then informs the user by displaying message error and suggest re-try.</td></tr>
</table>
</td></tr>
</table>

Table 12: Use case of sign-up

4.1.3 Functional Requirements

REQ-1: System should allow the user to enter first name, last name, E-mail, password, and birth date.

REQ-2: System must store the data provided by the user in the database.

REQ-3: System must display an plain message to indicate successful registration.

4.2 System Feature 2

Log-in

4.2.1 Description and Priority

Log-in allows users to securely access the HealthyCooking application to view and manage user recipes. The following figure illustrates the use-case of this feature. Priority of this feature is high. Benefit: 10 - Penalty: 8 - Cost: 3 - Risk: 5.

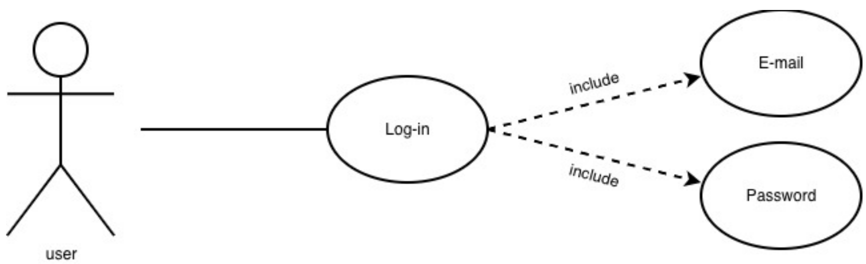

Figure 3: use case of the log-in feature

4.2.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Log-in</th></tr>
<tr><td>Goal</td><td>Enable user to log-in to the system.</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>Database</td></tr>
<tr><td>Pre-condition</td><td>User has an account in the application</td></tr>
<tr><td>Post-condition</td><td>User log-in to the system successfully.</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to log-in</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “log-in” button.</td></tr>
<tr><td>2</td><td>System displays E-mail and password filling interface.</td></tr>
<tr><td>3</td><td>User enters E-mail and password.</td></tr>
<tr><td>4</td><td>System checks the validity of the E-mail and password from database.</td></tr>
<tr><td>5</td><td>System shows a message of successful log-in.</td></tr>
<tr><td>6</td><td>System directs user to the home interface.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>4.1</td><td>The E-mail or password is not valid, system then informs the user by displaying message error and suggest re-try.</td></tr>
</table>
</td></tr>
</table>

Table 13: Use case of log-in

4.2.3 Functional Requirements

REQ-1: System should allow the user to enter E-mail and password.

REQ-2: System must validate the username and password against stored data.

REQ-3: System should grant access if the provided data are correct.

REQ-4: System must display an error message if the provided data are incorrect.

4.3 System Feature 3

Add Recipe

4.3.1 Description and Priority

Add recipe feature allows admins and users to add a new recipe to the HealthyCooking system database make it public to view to all users in the application. The following figure illustrates the use-case of this feature. Priority of this feature is moderate. Benefit: 6 - Penalty: 2 - Cost: 4 - Risk: 2.

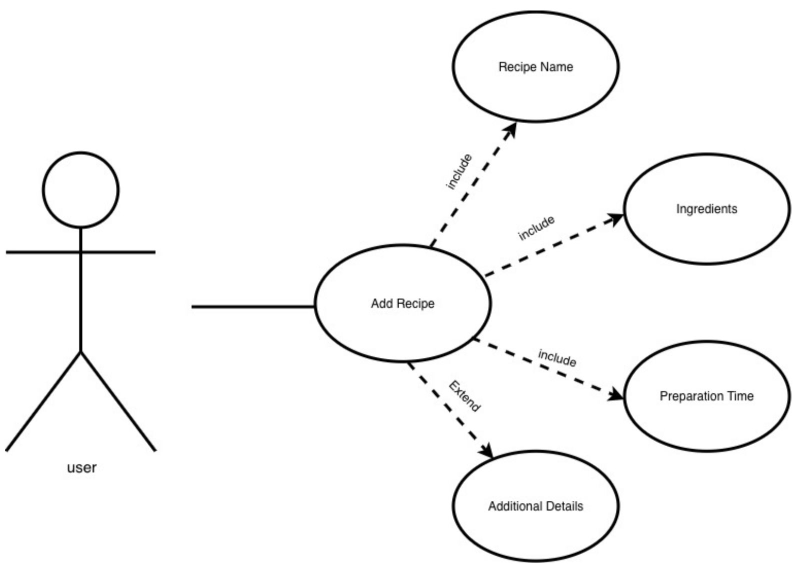

Figure 4: use case of the add recipe feature

4.3.2 Stimulus/Response Sequences

Use Case Name	**Add Recipe**
Goal	Enable user to add new recipe to the system.
Primary Actors	User
Secondary Actors	Database
Pre-condition	User has logged in to the system
Post-condition	User add recipe to the database successfully
Triggers	Use case starts when user indicates that they want to add new recipe
Main Flow	Step | Action 1 | User selects “add recipe” button. 2 | System displays recipe name, ingredients, preparation time, and additional details filling interface. 3 | User enters recipe name, ingredients, preparation time, and additional details. 4 | System store data and shows a message of successful recipe adding. 5 | System directs user to the home interface.
Extensions	Step | Action 3.1 | The recipe name or ingredients are not entered, system then informs the user by displaying message error and suggest re-try.

Main Flow:

Step	Action
1	User selects “add recipe” button.
2	System displays recipe name, ingredients, preparation time, and additional details filling interface.
3	User enters recipe name, ingredients, preparation time, and additional details.
4	System store data and shows a message of successful recipe adding.
5	System directs user to the home interface.

Extensions:

Step	Action
3.1	The recipe name or ingredients are not entered, system then informs the user by displaying message error and suggest re-try.

4.3.3 Functional Requirements

REQ-1: System should allow the user to all recipe information.

REQ-2: System must store the recipe information in the database.

REQ-3: System should display message to indicate successful recipe adding.

4.4 System Feature 4

Edit Recipe

4.4.1 Description and Priority

Edit recipe feature allows admins and users to edit existed recipe in the database. The following figure illustrates the use-case of this feature. Priority of this feature is low. Benefit: 3 - Penalty: 2 - Cost: 3 - Risk: 2.

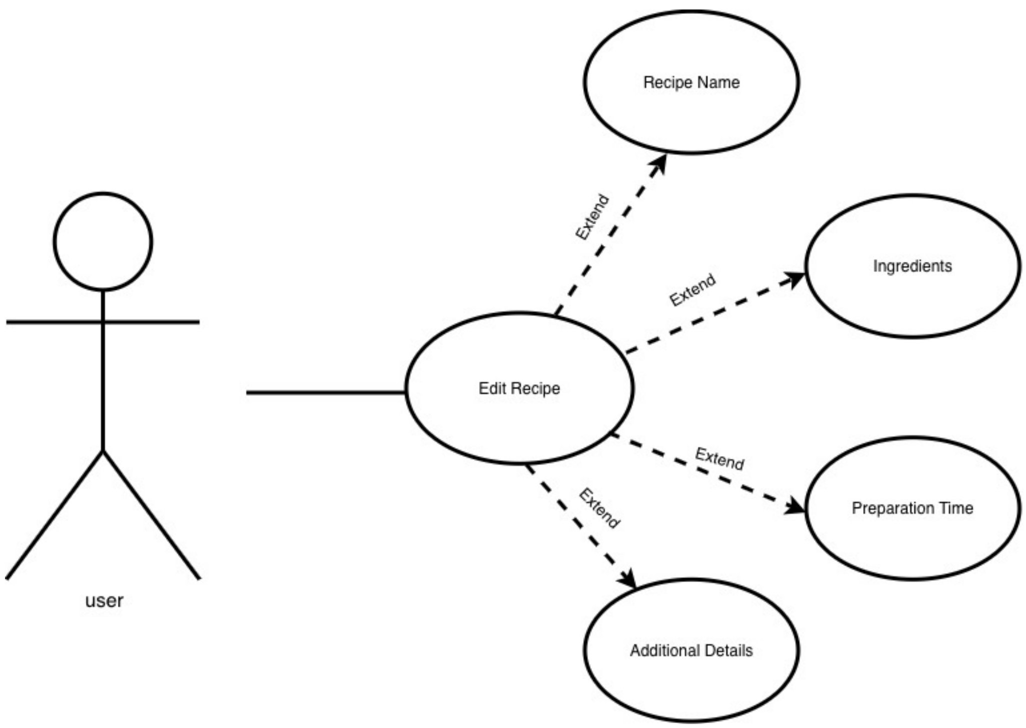

Figure 5: use case of the edit recipe feature

4.4.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Edit Recipe</th></tr>
<tr><td>Goal</td><td>Enable user to edit existing recipe in the system.</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>Database</td></tr>
<tr><td>Pre-condition</td><td>User has logged in to the system</td></tr>
<tr><td>Post-condition</td><td>User edit recipe in the database successfully</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to edit existing recipe</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “edit recipe” button.</td></tr>
<tr><td>2</td><td>System displays recipe names interface.</td></tr>
<tr><td>3</td><td>User choose recipe they want to edit.</td></tr>
<tr><td>4</td><td>System shows recipe details interface.</td></tr>
<tr><td>5</td><td>User edit the data of shown recipe.</td></tr>
<tr><td>6</td><td>System store entered data of recipe and show message to indicate successful recipe editing.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>5.1</td><td>The recipe name or ingredients are not entered, system then informs the user by displaying message error and suggest re-try.</td></tr>
</table>
</td></tr>
</table>

4.4.3 Functional Requirements

REQ-1: System should allow the user to chose one of the recipes to edit.

REQ-2: System must store the recipe information in the database.

REQ-3: System should display message to indicate successful recipe editing.

4.5 System Feature 5

Delete Recipe

4.5.1 Description and Priority

Delete recipe feature allows admins and users to delete existed recipe in the database. The following figure illustrates the use-case of this feature. Priority of this feature is moedrate. Benefit: 4 - Penalty: 6 - Cost: 3 - Risk: 4.

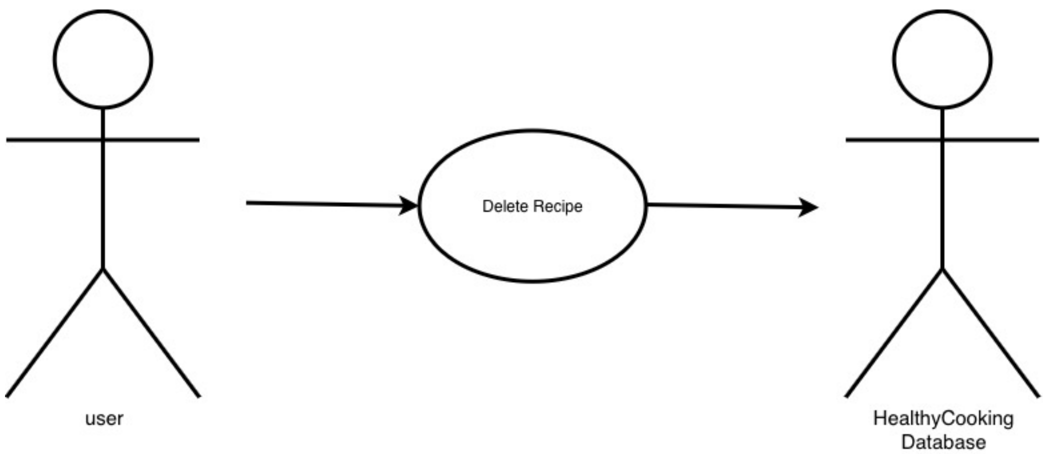

Figure 6: use case of the delete recipe feature

4.5.2 Stimulus/Response Sequences

Use Case Name	**Delete Recipe**
Goal	Enable user to delete existing recipe in the system.
Primary Actors	User
Secondary Actors	Database
Pre-condition	User has logged in to the system
Post-condition	User delete recipe from the database successfully
Triggers	Use case starts when user indicates that they want to delete existing recipe
Main Flow	<table><tr><th>Step</th><th>Action</th></tr><tr><td>1</td><td>User selects “delete recipe” button.</td></tr><tr><td>2</td><td>System displays recipe names interface</td></tr><tr><td>3</td><td>User choose recipe they want to delete</td></tr><tr><td>6</td><td>System deletes chosen recipe and show message to indicate successful recipe deleting.</td></tr></table>

4.5.3 Functional Requirements

REQ-1: System should allow the user to chose one of the recipes to delete.

REQ-2: System must delete the recipe from the database.

REQ-3: System should display message to indicate successful recipe deleting.

4.6 System Feature 6

Search Recipe

4.6.1 Description and Priority

Search recipe feature allows users to search recipe from the database. The following figure illustrates the use-case of this feature. Priority of this feature is high. Benefit: 9 - Penalty: 5 - Cost: 5 - Risk: 4.

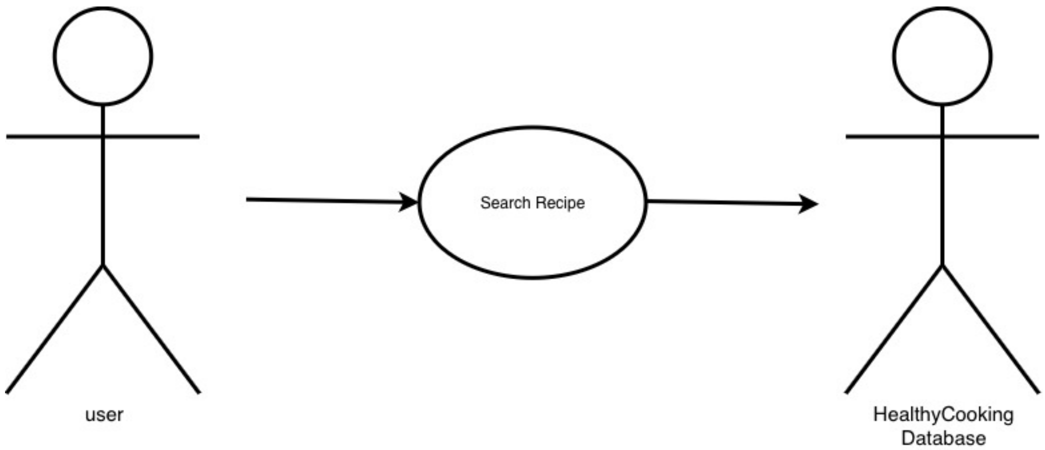

Figure 7: use case of the search recipe feature

4.6.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Search Recipe</th></tr>
<tr><td>Goal</td><td>Enable user to search existing recipes in the system</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>Database</td></tr>
<tr><td>Pre-condition</td><td>User has logged in to the system</td></tr>
<tr><td>Post-condition</td><td>User gets relevant results of recipe search</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to search recipe</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “search recipe” button.</td></tr>
<tr><td>2</td><td>System displays filling interface.</td></tr>
<tr><td>3</td><td>User enters recipe name, ingredients, or category they prefer.</td></tr>
<tr><td>4</td><td>System shows relevant recipes.</td></tr>
<tr><td>5</td><td>User choses recipe they prefer.</td></tr>
<tr><td>6</td><td>System shows chosen recipe details.</td></tr>
</table>
</td></tr>
</table>

4.6.3 Functional Requirements

REQ-1: System should allow the user to enter data to search for a recipe.

REQ-2: System must present relivent recipies from the database.

4.7 System Feature 7

Plan Weekly Meals

4.7.1 Description and Priority

Plan weekly meals feature allows users to plan meals for their week. The following figure illustrates the use-case of this feature. Priority of this feature is low. Benefit: 3 - Penalty: 3 - Cost: 7 - Risk: 5.

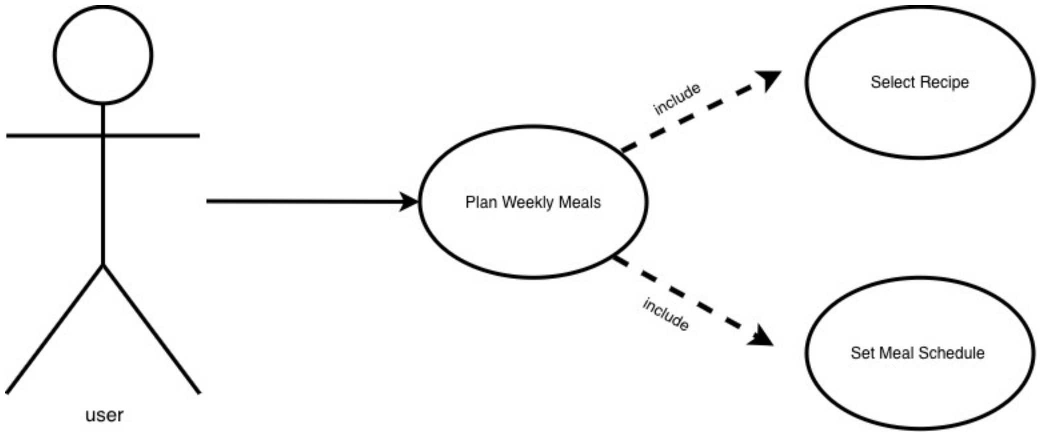

Figure 8: use case of the plan weekly meal feature

4.7.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Plan Weekly Meals</th></tr>
<tr><td>Goal</td><td>Enable user to plan their weekly meals with the system</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>Database</td></tr>
<tr><td>Pre-condition</td><td>User has logged in to the system</td></tr>
<tr><td>Post-condition</td><td>User plans their week diet successfully</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to plan weekly meals</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “plan week meals” button.</td></tr>
<tr><td>2</td><td>System displays recipe and time filling interface.</td></tr>
<tr><td>3</td><td>User enters recipe name from system, and schedule time.</td></tr>
<tr><td>4</td><td>System brings recipe ingredients and calculates recipe calories.</td></tr>
<tr><td>5</td><td>System shows message to indicate successful meal plan.</td></tr>
</table>
</td></tr>
</table>

4.7.3 Functional Requirements

REQ-1: System should allow the user to choose recipe to add to the weekly plan.

REQ-2: System should allow the user to set schdual for the recipe to add to the weekly plan.

4.8 System Feature 8

Track Calories

4.8.1 Description and Priority

Track calories feature allows users to track their calories throughout adding the meals they had in the system. The following figure illustrates the use-case of this feature. Priority of this feature is moderate. Benefit: 8 - Penalty: 6 - Cost: 4 - Risk: 8.

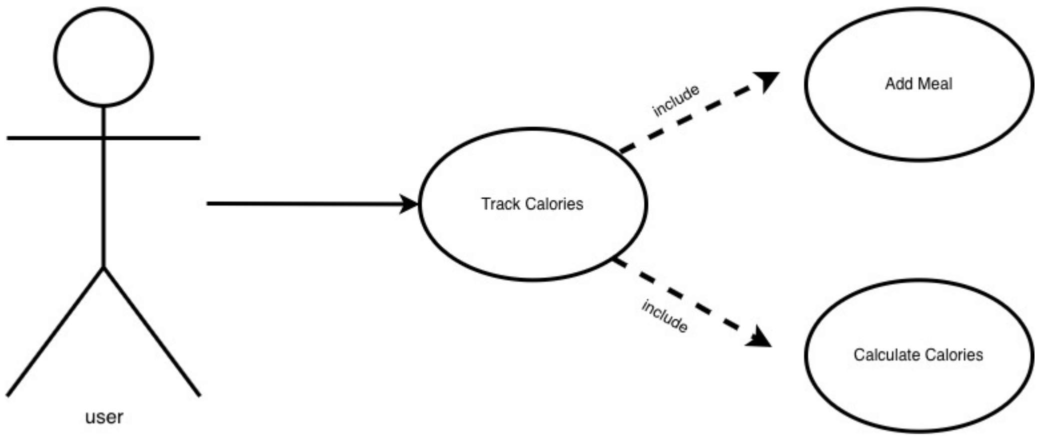

Figure 9: use case of the track calories feature

4.8.2 Stimulus/Response Sequences

Use Case Name	**Track Calories**
Goal	Enable user to track calories with the system.
Primary Actors	User
Secondary Actors	Database
Pre-condition	User has logged in to the system
Post-condition	User tracks their calories intake successfully
Triggers	Use case starts when user indicates that they want to track calorie
Main Flow	(see table below)
Extensions	(see table below)

Main Flow

Step	Action
1	User selects "track calories" button
2	System displays recipe name and ingredients filling interface
3	User enters recipe name and ingredients
4	System calculates calories of recipe ingredients
5	System store calculated data of recipe and show message to indicate successful calories tracking.

Extensions

Step	Action
3.1	The recipe name or ingredients are not entered, system then informs the user by displaying message error and suggest re-try.

4.8.3 Functional Requirements

REQ-1: System should allow the user to choose recipe or enter ingrediant.

REQ-2: System should calculates the calories of user input.

REQ-3: System should store calculated calories.

4.9 System Feature

Profilee Management

4.9.1 Description and Priority

Pprofile management feature allows users to set dietary objectives and add their allergies ingrediants in the systme. The following figure illustrates the use-case of this feature. Priority of this feature is high. Benefit: 8 - Penalty: 6 - Cost: 6 - Risk: 3.

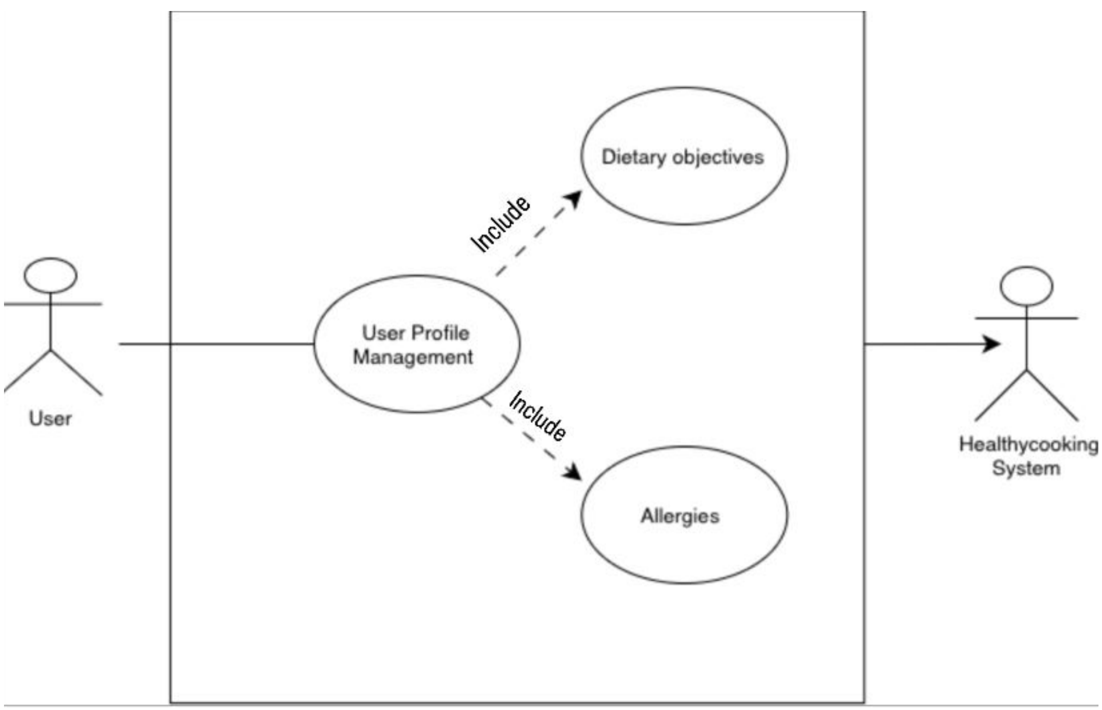

Figure 10: use case of user profile management feature

4.9.2 Stimulus/Response Sequences

Use Case Name	**Profile Management**
Goal	Enable user to add dietary and allergies information in the system
Primary Actors	User
Secondary Actors	Database
Pre-condition	User has logged in to the system
Post-condition	User adds their dietary information successfully
Triggers	Use case starts when user indicates that they want to manage profile
Main Flow	<table><tr><th>Step</th><th>Action</th></tr><tr><td>1</td><td>User selects “manage profile” button</td></tr><tr><td>2</td><td>System displays dietary objectives and allergies ingredients filling interface</td></tr><tr><td>3</td><td>User enters dietary objectives and allergies ingredients</td></tr><tr><td>4</td><td>System store entered data of recipe and show message to indicate successful data updating.</td></tr></table>

4.9.3 Functional Requirements

REQ-1: System should allow the user to enter dietary objectives and allergies ingredients.

REQ-2: System should store entered datasuccessfully.

4.10 System Feature

Personalized meal recommendation

4.10.1 Description and Priority

Personalized meal recommendation feature helps users finding suitable meals according to their diet and allergy information with the help of AI. The following figure illustrates the use-case of this feature. Priority of this feature is moderate. Benefit: 7 - Penalty: 2 - Cost: 6 - Risk: 3.

Figure 11: use case of personalized meal recommendation feature

4.10.2 Stimulus/Response Sequences

<table>
<tr><th>Use Case Name</th><th>Personalized meal recommendation</th></tr>
<tr><td>Goal</td><td>Help user find suitable meal according to their diet</td></tr>
<tr><td>Primary Actors</td><td>User</td></tr>
<tr><td>Secondary Actors</td><td>AI Generator</td></tr>
<tr><td>Pre-condition</td><td>User has logged in to the system</td></tr>
<tr><td>Post-condition</td><td>User finds suitable meal according to their data preferences</td></tr>
<tr><td>Triggers</td><td>Use case starts when user indicates that they want to have a meal recommendation</td></tr>
<tr><td>Main Flow</td><td>
<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1</td><td>User selects “meal recommendation” button</td></tr>
<tr><td>2</td><td>System AI displays recipe recommended names based on user stored data</td></tr>
<tr><td>3</td><td>User chooses meal they prefer</td></tr>
<tr><td>4</td><td>System shows meal information to the user</td></tr>
</table>
</td></tr>
</table>

4.10.3 Functional Requirements

REQ-1: System should search smartly using AI to give user meals recommendation.

REQ-2: System should search for meals based on user stored data.

4.11 System Feature

Meal Remainders

4.4.1 Description and Priority

Meal remainders feature remaind users of their meal time. The following figure illustrates the use-case of this feature. Priority of this feature is low. Benefit: 3 - Penalty: 2 - Cost: 3 - Risk: 2.

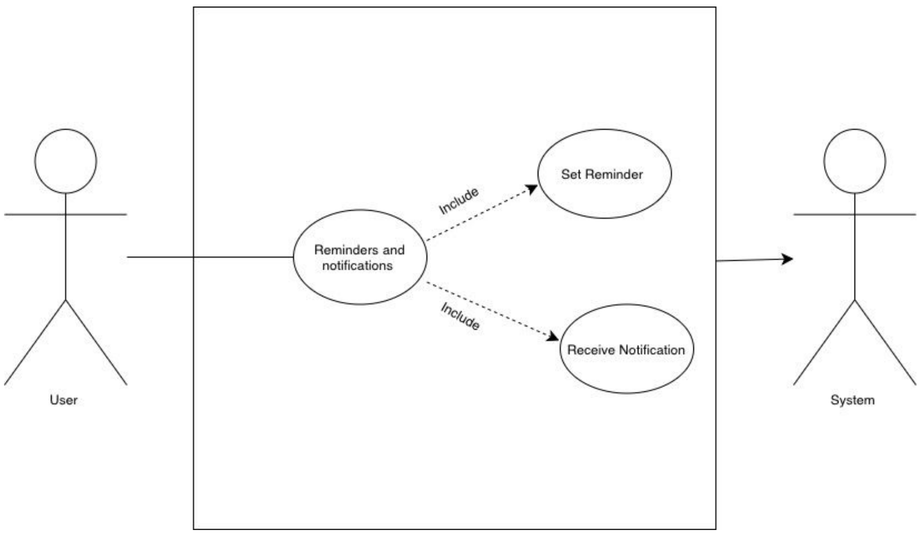

Figure 12: use case of the meal remainders feature

4.4.2 Stimulus/Response Sequences

Use Case Name	**Meal Remainders**
Goal	Remained users of their meals in time
Primary Actors	User
Secondary Actors	Database
Pre-condition	User has logged in to the system
Post-condition	User is remanded by their meal in time
Triggers	Use case starts when user indicates that they want to set remainder
Main Flow	<table><tr><th>Step</th><th>Action</th></tr><tr><td>1</td><td>User selects “set meal remainder” button.</td></tr><tr><td>2</td><td>System displays recipe names and notification time filling interface.</td></tr><tr><td>3</td><td>User chooses recipe name and enters time</td></tr><tr><td>4</td><td>System stores entered data and show message to indicate successful recipe editing.</td></tr></table>

4.4.3 Functional Requirements

REQ-1: System should allow the user to chosse recipe and set remainder time.

REQ-2: System must store the recipe and time information in the database.

REQ-3: System should display message to indicate successful data storing.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

HealthyCooking application is kept in a healthy state to run quickly and efficiently. It has to boot within a few seconds and its navigation between screens needs to be fast. In the standard conditions of the network and the equipment, the display of results should be provided in 1-3 seconds during the search of recipes by users or the refinement of the results. It is also necessary to ensure that the AI-based recommendations do not take an excessively long period to load so that they can offer a positive user experience.

The application must not take up much memory of the device or battery since this will lead to lagging or overheating. It should be reactive as recipes are browsed, nutrition information is read or custom weekly meals are changed. The system should also have the capability of handling numerous users simultaneously without interfering with performance of the system..

5.2 Safety Requirements

At Healthy Cooking application, we focus on security while guaranteeing quality and user safety A backup system is very important and is included in most applications, including ours, due to its importance. We want to protect all user information. We don't want to lose data or experience system malfunctions. We care deeply about our users because we are keen to prevent problems arising from inaccurate or misleading information, including food that may cause allergies or be harmful and unsuitable for some users All information and data presented in this program are verified and sourced from reliable food sources to avoid errors and problems. The app also include a disclaimer stating, "This program is not a substitute for consulting a doctor it is a nutritional aid and guide." This disclaimer is accompanied by a message confirming this The program also checks user profiles to identify foods that cause allergies or are not included in the user's account, preventing their addition or inclusion in certain recipes that may appear in the future. If the user is unaware of an increase in their calorie intake, the system will notify them.

5.3 Security Requirements

The protection and security of data includes a proposed security framework for user information and data within the application. This also includes system breaches, unauthorized access by any person, and misuse Everything related to the user and included in our application, from personal information and data requested when starting to use and registering for the program, to favorites and food preferences, is encrypted. We want to ensure its confidentiality and security. This is done through a specific algorithm that includes HTTPS protocols connecting the user and the server It is very important that we have put in place a system or mechanism to verify the information entered by the user and require the entry of a strong password under specific conditions. This is a basic policy that we agreed to put in place for security, because we do not want unauthorized access to information.

5.4 Software Quality Attributes

The following software quality attributes defined in this section contribute to the application.

Availability: a critical non-functional requirement of Healthycooking application since the application must be available 24/7 for the use of all end-users.

Maintainability: an important non-functional requirement for Healthycooking application as it is easy to extend and fix, which encourages the application's uptake and use; therefore, the components of the system will be self-contained as much as possible.

Portability: Healthycooking application must be portable since it will run on both iOS and Android which are different environments.

Reliability: our program is made to work dependably and ability to function correctlv and consistently over time. The system should maintain at least 99% uptime and recover automatically from minor failures.

Scalability: the system must be able to accommodate future expansion enabling the addition of additional meal plans recipes and features without adversely affecting overall performance. In order to meet growing user demands the system architecture will be developed.

Accessibility: is the ability for all users including those with physical or visual impairments to use the system efficiently. To guarantee legible text appropriate color contrast and simple assistive technology navigation global accessibility standards will be implemented.

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5.5 Business Rules

The following business rules define how the system will operate and what actions are permitted for different user roles:

- Rule 1: Only individuals with an account are permitted to develop or alter their own meal plans.
- Rule 2: The nutritional information must be presented in kilocalories and grams.
- Rule 3: The system will refrain from recommending recipes that contain ingredients to which the user is allergic.
- Rule 4: The recommended recipes must align with the user’s preferences and dietary objectives.
- Rule 5: Users have the ability to modify or remove their information at any moment to maintain privacy and accuracy.

These business rules ensure consistent behavior and compliance with the project's usability and ethical guidelines.

6. Other Requirements

The application includes additional requirements to support its functionality. It supports two languages Arabic, and English, to make it easier to use for a wider range of users. It also includes a backup and recovery feature to ensure that user data is safely stored. In addition, it provides data synchronization to keep information up to date across all devices signed in with the same account. Furthermore, the app supports cloud storage to securely save data and make it easily accessible.

Appendix A: Glossary

The acronyms and abbreviations in the HealthyCooking SRS document are defined in the following table. This ensures that all document participants and readers have a consistent understanding of the terminology used throughout the document.

Acronym / Term	Definition
AI	Artificial Intelligence: Algorithms that learn and generate intelligent recommendations.
API	Application Programming Interface: Channel that enables communication between software components.
SRS	Software Requirements Specification – a document that describes all functional and non-functional requirements of the system.
BMI	Is a number calculated from a user's height and weight to determine health and body-fat levels.
IEEE	Institute of Electrical and Electronics Engineers: Developer of key software engineering standards.
Admin	An admin is a system administrator with proper permissions to manage content and ensure data integrity.
User	A user is a registered person using the HealthyCooking app to monitor nutrition, manage meal plans, and explore recipes.
UI/UX	User Interface and User Experience: Design principles ensuring clarity and usability.
Recipe Database	This system stores of healthy cooking effective recipes. The recipe database is an industry strength database for storing validated. Which is user submitted recipe do.

Meal Planner	This feature allows the user to create and modify meal plans for the whole week according to the meals one prefers and the health needs of the customer.
WHO	World Health Organization, which is a global authority on health whose nutrition guidelines can be referenced in the system.
Smart Algorithms	Personalized recommendation engines based on intelligent algorithms offer advanced suggestions amalgamated with the user's lifestyle and eating habits.
Dietary Preferences	Dietary preferences of user including allergies, disliked ingredients, meal categories, or total calorie's goal.
Scalability	Scalability refers to system capabilities of expanding data, users, features, and more while maintaining performance and reliability.
HTTPS	Hypertext Transfer Protocol Secure, it is a secure version of the HTTP protocol, that uses encryption to protect the data transmitted between a user's device and the server.
PDF	A Portable Document Format, a file format developed by Adobe to present documents in a manner that is independent of software, hardware, and operating systems.
SDS	Software Design Specification, a document that describes the data needed to design and implement the system.
iOS	Apple's company operating system that is used for iPhone and iPad devices.
FAQ	A list of common questions and answers that help users understand and solve basic issues about the system.

Table 14: Document Terminologies

Appendix B: Analysis Models

This following UML class figure represents the main structure of the HealthyCooking application and how its components interact. It shows the key classes such as User, Recipe, Ingredient, MealPlan, NutritionInfo, Review, Admin, and RecommendationEngine, along with their attributes and main functions.

The relationships between these classes define how data flows through the system. For example, users can create reviews and meal plans, recipes are linked to ingredients and nutritional information, and the recommendation engine connects user preferences with suitable recipes.

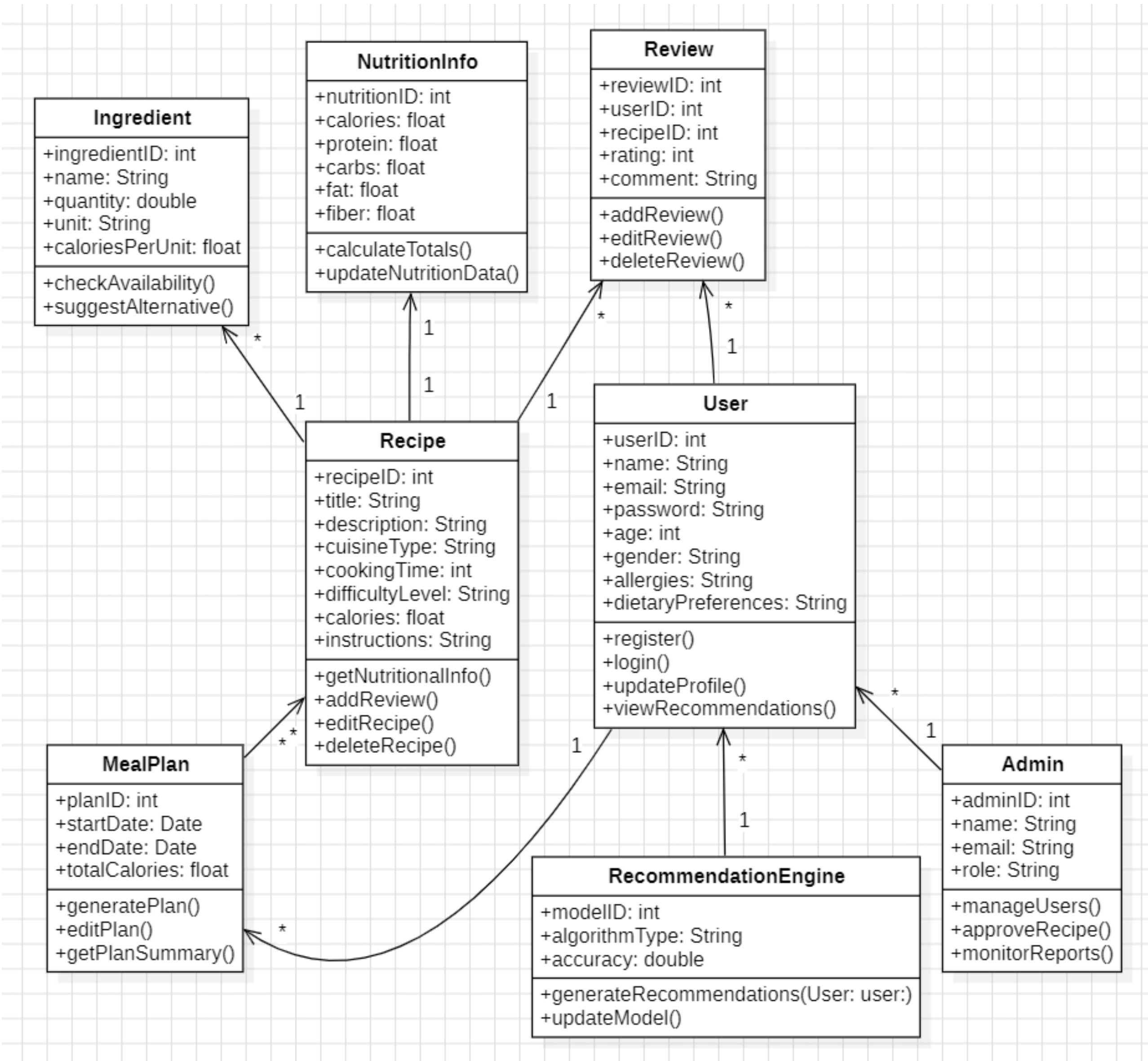

Figure 13: UML class diagram representing the main structure of the app

Appendix C: To Be Determined List

- The final database system that will be used.
- The hosting server or cloud provider.
- The specific data security certification or privacy compliance standard.
- The final release platforms (iOS only or iOS and Android) are not fully confirmed.
- The complete user interface design, including colors and icons, will be finalized in later SDS stages.